Program : **Diploma in Cloud Computing and Big Data**	
Course Code: **3272**	Course Title: **Database System Concepts**
Semester : **3**	Credits: **3**
Course Category: **Program Core**	
Periods per week: **3 (L:3 T:0 P:0)**	Periods per semester: **45**

Course Objectives:

- Impart the basic concepts of Database Management Systems.
- Enable students to construct relational database design using ER Model and Normalization techniques
- Provide a solid technical overview of SQL.
- Give an overview of transaction processing, concurrency control and NOSQL databases.

Course Prerequisites:

Topic	Course code	Course name	Semester
Computer Fundamentals	1008	Introduction to IT systems Lab	I
Programming concepts	2131	Problem Solving & Programming	II

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Describe Database Management System Concepts and Relational Data Model.	10	Understanding
CO2	Demonstrate building and manipulating databases using Structured Query Language.	11	Applying
CO3	Illustrate the Relational Database Design using ER Model and Normalization	11	Applying

CO4	Explore the real time transaction processing and NOSQL database	11	Understanding
	Series Test	2	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	-	1	-	-	-	-
CO2	3	3	3	2	-	-	-
CO3	3	3	3	-	-	-	-
CO4	3	3	3	2	-	-	-

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Name of the Experiment	Duration (Hours)	Cognitive Level
CO1	**Describe Database Management System Concepts and Relational Data Model**		
M1.01	Outline the concept of database and database management system.	3	Understanding
M1.02	Explain Database Architecture and Languages.	3	Understanding
M1.03	Describe Basic Relational Model Concepts	4	Understanding

Contents:

Introduction to Databases: Definition of Data, Database, and DBMS, Characteristics of Database Approach, Advantages of DBMS, Applications of Database systems, Different types of individuals interacting with DBMS, Data Models, Database Schema - Instance - Database state, Three Schema Architecture and Data Independence, Database Languages, Database Interfaces, DBMS Component Modules, Centralized and Client/Server Architecture, Classification of DBMS.

Relational Model Concepts: Domain, Attributes, Tuples and Relations, Characteristics of Relations, Relational Model Constraints and Relational Database Schemas, Update Operations and Dealing with Constraint Violations.

<table>
<tr><td>CO2</td><td colspan="3">Demonstrate building and manipulating databases using Structured Query Language.</td></tr>
<tr><td>M2.01</td><td>Construct basic queries for data definition, retrieval and modification using SQL</td><td>5</td><td>Applying</td></tr>
<tr><td>M2.02</td><td>Implement complex data retrieval queries, assertions, triggers and schema change statements using SQL</td><td>6</td><td>Applying</td></tr>
<tr><td></td><td>Series Test – I</td><td>1</td><td></td></tr>
<tr><td colspan="4">Contents:
SQL: Data Definition and Data Types, Specifying Constraints, Basic Retrieval Queries, Insert, Delete and Update Statements.
Advanced Features of SQL: Complex SQL Retrieval Queries, Joins, Aggregate functions and grouping, Assertions, Triggers, Views, Schema change statements.</td></tr>
<tr><td>CO3</td><td colspan="3">Illustrate Relational Database Design using ER Model and Normalization.</td></tr>
<tr><td>M3.01</td><td>Describe the phases of database Design</td><td>1</td><td>Understanding</td></tr>
<tr><td>M3.02</td><td>Explain Entity Relationship Model.</td><td>2</td><td>Understanding</td></tr>
<tr><td>M3.03</td><td>Extend Enhanced ER Model.</td><td>1</td><td>Understanding</td></tr>
<tr><td>M3.04</td><td>Construct ER Models for real world applications.</td><td>3</td><td>Applying</td></tr>
<tr><td>M3.05</td><td>Outline the informal design guidelines for relation schema</td><td>1</td><td>Understanding</td></tr>
<tr><td>M3.06</td><td>Illustrate normalization using Functional dependency - 1NF - 2NF - 3NF - BCNF - 4NF</td><td>3</td><td>Applying</td></tr>
<tr><td colspan="4">Contents:
Conceptual Modelling: Phases of Database Design, Definition-Entity Types, Entity Sets, Attributes and Keys, Relationship Types, Relationship sets, Roles, Structural Constraints, Weak Entity Types,
ER Diagrams, Naming Conventions, Relationship Types of Degree higher than two. Design ER model for Real applications.
Enhanced ER model: Specialization and Generalization
Functional Dependency and Normalization: Informal design guidelines for relation schema, Functional Dependencies, Normal Forms – 1NF, 2NF, 3NF, BCNF, 4NF.</td></tr>
<tr><td>CO4</td><td colspan="3">Explore the real time transaction processing and NOSQL database</td></tr>
</table>

M4.01	Describe the need for Concurrency Control and recovery	2	Understanding
M4.02	Explain the desirable properties of Transaction and states of transaction execution	2	Understanding
M4.03	Extend the characteristics of NOSQL databases.	1	Understanding
M4.04	Outline the CAP Theorem	1	Understanding
M4.05	Classify the NOSQL databases.	2	Understanding
M4.06	Demonstrate the Document-Based NOSQL Systems - MongoDB	3	Applying
	Series Test – II	1	

Contents:

Transaction Processing: Introduction to Transaction Processing, Need for Concurrency Control and Recovery, Transaction states, Desirable Properties of Transactions.

NOSQL Databases: Introduction to NOSQL, Characteristics of NOSQL systems, Desirable properties of distributed systems with replicated data, CAP Theorem, Categories of NOSQL systems :

Document-Based NOSQL Systems - MongoDB,

NOSQL Key-Value Stores - DynamoDB*,

Column-Based NOSQL Systems - Hbase Data model and versioning*,

NOSQL Graph Databases - Neo4j Data Model*

*Basic concepts only

Text / Reference

T/R	Book Title/Author
T1	Elmasri, Navathe, **Fundamentals of Database Systems**, Pearson, 7th Ed.
R1	Silberschatz, Korth, and Sudarshan, **Database system concepts**, McGrawHill, 6th Ed.
R2	Bayross, Ivan BPB, **SQL/ PL/SQL**, New Delhi

Online Resources

Sl.No	Website Link
1	https://nptel.ac.in/courses/106106093/
2	https://www.w3schools.com/sql
3	https://www.tutorialspoint.com/sql
4	https://www.javatpoint.com/sql
5	https://www.javatpoint.com/mongodb-tutorial
6	https://docs.mongodb.com/manual/tutorial/getting-started/